

Real-time PCR to target *Hoodia* in herbal supplements: a tool for conservation and trade regulation

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ABSTRACT

Plants belonging to the *Hoodia* genus are widely used as ingredients in herbal dietary supplements. Rising demand and overharvesting have led to supply shortages, adulteration, and CITES-regulated trade, threatening the authenticity and safety of *Hoodia* in commercial products. These factors highlight the need for a reliable genus-level detection method for *Hoodia* to support the control of illegal trade and the verification of marketed *Hoodia*-based products. To address this need, we developed a real-time PCR method specific to *Hoodia* genus targeting the ITS2 region, a molecular marker with high variability enabling detection and differentiation of plant species or genera. Due to the limited number of publicly available *Hoodia* ITS2 sequences, 13 additional sequences were generated in-house to support assay design. This real-time PCR method was validated in compliance with international standards, demonstrating high specificity, high sensitivity, and inter-laboratory transferability. Its applicability was verified by testing commercial herbal dietary supplements. This study provides enforcement laboratories with a validated molecular tool to support *Hoodia* monitoring and authentication in commercial products.

1. Introduction

The *Hoodia* genus, part of the *Apocynaceae* family, consists of spiny, leafless succulent plants native to Southern Africa. It has drawn attention for its potential appetite suppressant properties, which are associated with the presence of the oxypregnane glycoside P57AS3 (Avula et al., 2006, 2008; MacLean & Luo, 2004; Ríos-Hoyo & Gutiérrez-Salmeán, 2016; Rumalla et al., 2008; Smith & Krygsman, 2014; Van Heerden, 2008; Vermaak et al., 2011; Wang, Xiao, et al., 2023; Zhao et al., 2011). Consequently, the demand for *Hoodia* species, such as *H. gordonii*, *H. currorii*, *H. parviflora* and *H. ruschii*, has significantly increased, particularly due to their commercial use in herbal dietary supplements (Avula et al., 2008; Avula et al., 2006; Bonetti et al., 2022; Gathier et al., 2013; Joshi et al., 2009; Ríos-Hoyo & Gutiérrez-Salmeán, 2016; Smith & Krygsman, 2014; Van Heerden, 2008; Vermaak et al., 2011; Zhao et al., 2011). This surge in demand has led to excessive harvesting of this slow-growing succulent plant genus, thereby raising

biodiversity conservation concerns. Therefore, the *Hoodia* genus was listed under Appendix II of the Convention on International Trade in Endangered Species of Wild Fauna and Flora (CITES). This classification, which remains in force, mandates permits for the cultivation and export of both wild and artificially propagated materials (Avula et al., 2008; Bonetti et al., 2022; CITES, 2006; CITES, 2025; Joshi et al., 2009; Smith & Krygsman, 2014; Van Heerden, 2008; Vermaak et al., 2011; Zhao et al., 2011). Overexploitation of *Hoodia* has additionally led to supply shortages and widespread adulteration with other plant materials. For instance, *H. gordonii* has frequently been substituted with *Opuntia ficus indica*, and many marketed *Hoodia*-based supplements or raw materials have been found to contain only little or no *Hoodia*. Such practices raise potential concerns about the authenticity, quality, traceability and safety of the commercial *Hoodia*-based products (Avula et al., 2008; Joshi et al., 2009; Koncz et al., 2021; Smith & Krygsman, 2014; Van Heerden, 2008; Vermaak et al., 2011; Zhao et al., 2011). In addition, potential public health concerns have arisen because clinical studies

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failed to demonstrate significant *Hoodia* weight-loss effects, while adverse cardiovascular and neurological effects have been reported (Avula et al., 2006; Blom et al., 2011; Le Nevé et al., 2010; Madgula et al., 2010; Ríos-Hoyo & Gutiérrez-Salmeán, 2016; Smith & Krygsman, 2014; Van Heerden, 2008; Zhao et al., 2011). For these reasons, several countries have banned the use of *Hoodia* in food supplements. On the European market, *Hoodia*-containing supplements are classified as novel food and cannot legally be marketed without prior authorization. Nevertheless, non-compliant products are frequently seized by customs authorities (Koncz et al., 2021).

Collectively, these challenges have consequently highlighted the need for a reliable genus-level detection method for *Hoodia* to enforce regulatory compliance and to monitor illegal trade and product authenticity, thereby contributing to consumer protection (Avula et al., 2006; Avula et al., 2017, 2008; CITES, 2006; Coutinho Moraes et al., 2015; Gathier et al., 2013; Janssen et al., 2008; Joshi et al., 2009; Rojas et al., 2011; Rumalla et al., 2008; Zhao et al., 2011). Although official methods and standardized protocols are currently lacking across enforcement laboratories, different approaches have been explored. These include macroscopy and microscopy, which are, however, limited by the highly processed nature of herbal ingredients in commercial supplements and the specialized expertise required (Coutinho Moraes et al., 2015; Gathier et al., 2013; Joshi et al., 2009). Consequently, alternative approaches have been investigated, including predominantly chemical-based methods, as well as DNA-based methods (Avula et al., 2006; Avula et al., 2017, 2008; Gathier et al., 2013; Janssen et al., 2008; Joshi et al., 2009; Rumalla et al., 2008; Zhao et al., 2011). However, chemical profiling is influenced by plant tissue types, developmental stages and environmental conditions (Avula et al., 2006; Avula et al., 2017, 2008; Coutinho Moraes et al., 2015; Gathier et al., 2013; Janssen et al., 2008; Joshi et al., 2009; Rojas et al., 2011; Rumalla et al., 2008; Zhao et al., 2011). In contrast, DNA-based methods offer a reliable and powerful complementary alternative, as DNA is a stable biological molecule found in all tissues, regardless of developmental stage or environmental factor, and often persists in processed materials. Nevertheless, only a few studies have applied DNA-based methods to detect *Hoodia*, including conventional PCR followed by amplicon size discrimination via electrophoresis or conventional PCR followed by Sanger sequencing (Coutinho Moraes et al., 2015; Gathier et al., 2013; Joshi et al., 2009; Rojas et al., 2011). However, these DNA-based methods present notable limitations for enforcement laboratories. Electrophoresis-based approaches are often avoided due to limited precision, sensitivity, resolution, and dynamic range. Sequencing-based approaches are not commonly employed because they require access to sequencing infrastructure and advanced bioinformatics expertise. The presence of mixed plant species in the tested sample, as commonly found in herbal supplements, further complicates the analysis, making the process time-consuming and technically challenging (Gathier et al., 2013; Joshi et al., 2009; Prigigallo et al., 2016). In contrast to these DNA-based methods, real-time PCR is a powerful, user-friendly, time-efficient alternative widely regarded as the gold standard by enforcement laboratories. This technology enables real-time monitoring of DNA amplification, eliminating post-PCR processing and accelerating data acquisition. Importantly, it can reliably detect specific target sequences even within complex mixtures of multiple species, a common challenge in herbal weight-loss products. Moreover, its ability to amplify short DNA fragments is particularly valuable for analyzing degraded DNA typically found in highly processed materials, such as commercial *Hoodia* ingredients (Coutinho Moraes et al., 2015; Gathier et al., 2013; Joshi et al., 2009). The successful use of real-time PCR has been demonstrated in various fields. These include food safety, traceability, authenticity and fraud detection, but also illegal trade monitoring of endangered animal species in food products (Cardenosa et al., 2018; Deliveyne et al., 2022; Fraiture et al., 2015; Fraiture, Gobbo, Guillitte, Barhdadi, et al., 2024; Henger et al., 2023; Kang, 2019; Rojas et al., 2011; Villanueva-Zayas et al., 2021). In herbal-based supplements, real-

time PCR has also been used to detect and authenticate plant species of interest such as *Ginkgo biloba*, *Centella asiatica*, *Panax notoginseng*, and *Carica papaya* (Biltes et al., 2025; Grazina et al., 2020; Lou et al., 2022; Patel et al., 2023). Among the genomic regions widely used in real-time PCR strategies for plant authentication, the multi-copy internal transcribed spacer 2 (ITS2) region of the nuclear ribosomal DNA has emerged as a key molecular marker (Grazina et al., 2020; Lou et al., 2022; Patel et al., 2023; Tao et al., 2025). ITS2 is characterized by high interspecific sequence variability while remaining relatively conserved within species, providing strong discriminatory power for the reliable detection and discrimination of closely related plant species or genera (Banchi et al., 2020; Frigerio et al., 2021; Gu et al., 2013; Yao et al., 2010). Despite the many advantages of real-time PCR technology and the demonstrated suitability of the ITS2 marker for plant authentication, no real-time PCR assay, including those targeting ITS2, has to date been developed for *Hoodia* authentication, likely in part due to the limited DNA sequence data available for *Hoodia* and related species, which represents a critical constraint for assay design (Gathier et al., 2013; Joshi et al., 2009).

Therefore, to support enforcement laboratories in controlling marketed *Hoodia*-based products, a new real-time PCR method specific to the *Hoodia* genus was developed in this study by targeting the well-established ITS2 molecular marker. Because publicly available *Hoodia* ITS2 sequences were limited, 13 additional sequences were generated in-house to support real-time PCR assay design. The developed real-time PCR assay was validated in accordance with European standards, including the minimum performance requirements (MPR) for GMO analysis defined by the European Network of GMO Laboratories (ENGL), enabling a harmonized and reliable application across enforcement laboratories (Marchesi et al., 2015). First, specificity was assessed using DNA from several *Hoodia* species, closely-related species and plant species frequently encountered in commercial herbal weight-loss products. Genus-level detection was intentionally targeted because the entire *Hoodia* genus is listed under Appendix II of CITES, adulteration can affect all *Hoodia*-based products, and regulatory measures in many jurisdictions apply broadly to the genus (Avula et al., 2006; Avula et al., 2008; Bonetti et al., 2022; CITES, 2025; Joshi et al., 2009; Koncz et al., 2021; Smith & Krygsman, 2014; Van Heerden, 2008; Vermaak et al., 2011; Zhao et al., 2011). Second, sensitivity was evaluated by determining the limit of detection through serial dilutions of the targeted *Hoodia* sequence. Third, the transferability of the in-house validated *Hoodia* real-time PCR method to an external laboratory was evaluated. Finally, its applicability was tested using commercial herbal weight-loss products labelled or suspected to contain *Hoodia* ingredients.

2. Materials and methods

2.1. Biological materials

Plant materials were collected from various sources, including Amsbio (UK), American Herbal Pharmacopoeia (USA), ANSM (France), AOCS (USA), BOTANICERT® (France), Cambridge botanical garden (UK), CRA-W (Belgium), Fagron (Belgium), Giromagi (Italy), JRC (EU), Lyon botanical garden (France), Meise botanical garden (Belgium), Rarepalmseeds (Germany) and Ugent botanical garden (Belgium) (Tables S1–2). First, a total of 13 different *Hoodia* materials, representing 7 different *Hoodia* species, were included. These plant materials comprised 2 specimens of *H. currorrii*, 4 specimens of *H. gordonii*, 3 specimens of *H. parviflora* and one specimen of *H. juttae*, *H. pedicellate*, *H. pilifera* and *H. ruschii* (Table S1). Second, 17 samples from plant species closely related to *Hoodia* were collected, including *Caralluma europea*, *C. europaea* var. *Europea*, *C. joannis*, *C. hesperidum*, *Ceropegia woodii*, *Dischidia nummularia*, *D. ruscifolia*, *Dischidia* sp., *Hoya multiflora*, *Huernia keniensis*, *H. rubra*, *Orbea variegata*, *Rauvolfia vomitoria*, *Senna alexandrina*, *Stapelia flaviviridis/grandiflora*, *S. tromotriche revoluta* and *Strophanthus amboensis/petersianus* (Table S2). Among these, *Caralluma*

is a noteworthy plant genus for being commonly used alongside *Hoodia* ingredients in herbal weight-loss food products (Cohen et al., 2023; Corley & Miller, 2002; de Freitas Junior & de Almeida, 2017; Dutt et al., 2012; Elvir-Lazo et al., 2024; Gamboa-Gómez et al., 2015; König, 2024; Pittler et al., 2005; Sirotkin, 2022; Sun et al., 2021). Third, 8 samples from plant species not closely related to *Hoodia* but well-known to be used with *Hoodia* ingredients in herbal weight-loss food products were collected. These included *Citrus aurantium* (bitter orange), *Capsicum annuum* (paprika), *Camelia sinensis* (tea plant), *Foeniculum vulgare* (fennel), *Garcinia cambogia/gummi-gutta*, *Ilex paraguariensis* (Yerba mate), *O. ficus indica* (Prickly pear cactus) and *Paullinia cupana* (Guarana) (Table S2) (Cohen et al., 2023; Corley & Miller, 2002; de Freitas Junior & de Almeida, 2017; Dutt et al., 2012; Elvir-Lazo et al., 2024; Gamboa-Gómez et al., 2015; König, 2024; Pittler et al., 2005; Sirotkin, 2022; Sun et al., 2021). Fourth, 2 samples belonging to plant species not closely related to *Hoodia* and frequently found in herbal weight-loss food products, but not specifically mixed with *Hoodia* ingredients, were included. These are *Carica papaya* (Papaya) and *Cynara scolymus* (Artichoke) (Table S2) (Cohen et al., 2023; Corley & Miller, 2002; de Freitas Junior & de Almeida, 2017; Dutt et al., 2012; Elvir-Lazo et al., 2024; Gamboa-Gómez et al., 2015; König, 2024; Pittler et al., 2005; Sirotkin, 2022; Sun et al., 2021). Finally, 10 samples belonging to plant species neither closely related to *Hoodia* nor commonly observed in herbal weight-loss food products were used, including *Beta vulgaris*, *Brassica napus*, *Glycine max*, *Gossypium hirsutum*, *Oryza sativa*, *Triticum aestivum*, *Zea mays*, *Nicotiana tabacum*, *Solanum lycopersicum* and *S. tuberosum* (Table S2).

2.2. DNA extraction

Fresh plant material was cut into small pieces using a surgical blade, and placed in an incubator at 65 °C until fully dried. This material was then ground into a fine powder using an analytical mill (IKA-Werke GmbH, Germany) and stored at −20 °C until further analysis.

DNA was extracted using NucleoSpin® Food Kit (MACHEREY-NAGEL, Dueren, Germany) according to manufacturer's instructions. DNA concentration was measured by fluorometry using Qubit 4.0 Fluorometer (Thermo Fisher Scientific, Waltham, USA) and DNA purity was evaluated based on the A260/A280 and A260/A230 ratios provided by spectrophotometry using Nanodrop® 2000 (Thermo Fisher Scientific, Waltham, USA). Extracted DNA was stored at −20 °C.

2.3. ITS2 barcoding

From each *Hoodia* DNA sample, a conventional PCR assay was applied using a standard 25 µl reaction volume containing 1× KAPA HiFi HotStart ReadyMix (Roche, Basel, Switzerland), 250 nM of each primer (Eurogentec, Liège, Belgium), and 25 ng of DNA. The primer pair (ITS2-F: ATGCGATACTTGTTGTGAAT; ITS2-R: GACGCTTCTCCAGACTACAAT) used in this study was previously designed (Gu et al., 2013). The PCR assays were performed on a T100™ Thermal Cycler (Bio-Rad, Hercules, USA). The PCR program consisted of a single cycle at 95 °C for 5 min (initial denaturation), 30 cycles at 95 °C for 30 s (denaturation), at 54 °C for 40 s (annealing), and at 72 °C for 60 s (extension), and a single cycle at 72 °C for 10 min (final extension). From each *Hoodia* material, the final PCR product was visualized by electrophoresis using TapeStation4200 with associated D1000 ScreenTapes and reagents (Agilent, Santa Clara, USA), under default parameters according to the manufacturer's instructions. The final PCR product was then purified using ExoSAP-IT PCR Product Cleanup (Affymetrix, Santa Clara, USA) and subsequently sequenced on a Genetic Sequencer 5000 (Thermo Fisher Scientific, Waltham, USA) using the Big Dye Terminator Kit v3.1 (Applied Biosystems, Waltham, USA).

2.4. In-silico real-time PCR design

2.4.1. In-silico inclusivity assessment

All publicly available sequences belonging to *Hoodia* and corresponding to the ITS2 region were collected from NCBI and vascular plant ITS2 databases (access on 28/05/2024) (National Center for Biotechnology Information (NCBI), 2024; Quaresma et al., 2024), in addition to the in-house ITS2 sequences generated from *Hoodia* materials. Following an alignment of these sequences using CLUSTAL OMEGA (V1.2.4) with default parameters, a conservative region for *Hoodia* was selected for primer and probe design (Tables S3A–B) (Sievers & Higgins, 2021). The resulting real-time PCR oligonucleotides, including a forward primer (Hoodia-F: GCGTCGCCTCCTGCATAA), a reverse primer (Hoodia-R: TGGCGAGCGACGAATCTTTTA) and a probe (Hoodia-P: FAM-TG + C + C + TT + G + G + TGC-3IABkFQ), were manually designed, using IDT OligoAnalyzer™ Tool for checking parameters (<https://eu.idtdna.com/calc/analyzer>) (Tables S3A–B). Key design parameters were evaluated, like oligonucleotide length, GC content, melting temperature (T_m), and potential secondary structures or dimers. The probe design included a custom TaqMan probe incorporating Locked Nucleic Acid (LNA) bases, indicated by “+” at specific positions, to enhance specific hybridization efficiency. Using CLUSTAL OMEGA (V1.2.4) with default parameters, the expected PCR amplicon sequence generated by the *Hoodia* real-time PCR method was aligned to all available *Hoodia* ITS2 region sequences collected from NCBI and vascular plant ITS2 databases, as well as generated in-house from *Hoodia* materials (Table S3C) (Sievers & Higgins, 2021).

2.4.2. In-silico exclusivity assessment

A BLAST search, using default parameters, of the expected PCR amplicon generated by the designed *Hoodia* real-time PCR method, was performed against the NCBI nucleotide collection (nr/nt) (Table S4A) (National Center for Biotechnology Information (NCBI), 2024). All resulting hits were aligned using CLUSTAL OMEGA (V1.2.4) with default parameters (Table S4B) (Sievers & Higgins, 2021). All observed single-nucleotide variations at the Hoodia-P probe sequence level were highlighted (Tables S4B–C). Therefore, in order to increase the discrimination between the targeted Hoodia-P probe sequence and the closely related sequences, two additional non-Hoodia probes (non-Hoodia-Pa: HEX-CG + TG + C + TTT + G + A + TGGC-3IABkFQ; non-Hoodia-Pb: HEX-CGTG+C + C + TT + G + A + TGC-3IABkFQ) were manually designed, using IDT OligoAnalyzer™ Tool for checking parameters (<https://eu.idtdna.com/calc/analyzer>), and subsequently integrated into the final oligonucleotide set of the developed real-time PCR method. The probe design included custom TaqMan probes incorporating LNA bases, indicated by “+” at specific positions.

2.5. Experimental real-time PCR performance assessment

The performance of the developed *Hoodia* real-time PCR method was experimentally assessed according an international standard guidance document (Marchesi et al., 2015).

For each real-time PCR reaction, a 25 µl standard volume was prepared, including 1× SsoAdvanced universal probes supermix (Bio-Rad, Hercules, USA), 400 nM of Hoodia-F and Hoodia-R primers (Eurogentec, Liège, Belgium), 80 nM of Hoodia-P probe (IDT, Leuven, Belgium), 600 nM of non-hoodia-Pa and non-Hoodia-Pb probes (IDT, Leuven, Belgium) and 5 µl of DNA. Preliminary tests were conducted to optimize probe concentrations, showing that a higher concentration of the non-target probes in combination with a lower concentration of the target probe enhanced assay specificity, ultimately resulting in no false-positive signals (Table S5).

The real-time PCR program consisted of an initial cycle for DNA polymerase activation at 95 °C for 10 min, followed by 45 amplification cycles of a 15 s denaturation step at 95 °C and a 60 s annealing-extension step at 64 °C. All real-time PCR runs were conducted using the CFX96

Touch Real-Time PCR Detection System (Bio-Rad, Hercules, USA) and included a No Template Control (NTC). A real-time PCR reaction was considered as negative if either no C_q value or a C_q value ≥ 38 was observed.

2.5.1. Specificity

The developed real-time PCR method was tested in duplicate on about 3000 estimated haploid genome copies from several target and non-target materials gathered from different collections (Tables S1–2) (Arumuganathan & Earle, 1991; Pellicer & Leitch, 2020). In addition to plant materials described in Section 2.1., a total of 4 non-plant DNA samples belonging to animal and microbial species were also tested, including *Homo sapiens* (G3041, Promega, USA), *Bacillus licheniformis* (LMG 7558, BCCM, Belgium), *Aspergillus niger* (IHEM 2312, BCCM, Belgium) and *Pichia pastoris* (MUCL 27793, BCCM, Belgium).

To verify its sequence identity, the PCR amplicon generated from the developed real-time PCR method applied on the *H. gordonii* (1) material was purified using USB ExoSAP-IT PCR Product Cleanup (Affymetrix, Santa Clara, USA) and sequenced on a Genetic Sequencer 5000 (Thermo Fisher Scientific, Waltham, USA) using the Big Dye Terminator Kit v3.1 (Applied Biosystems, Waltham, USA). The generated sequence was aligned against the expected *Hoodia* PCR amplicon sequence using CLUSTAL OMEGA (V1.2.4) with default parameters (Table S6) (Sievers & Higgins, 2021).

2.5.2. Sensitivity

A gBlocks Gene Fragment (IDT, Leuven, Belgium), artificially synthesized to contain a single copy of the expected PCR amplicon sequence produced by the developed *Hoodia* real-time PCR method, was utilized (Table S3C). Following the manufacturer's instructions, this synthetic control material was employed to prepare serial dilutions ranging from 25 to 0.1 estimated target copies and tested in 12 replicates. The limit of detection LOD_{95%} was calculated as previously described (Table S7) (Grohmann et al., 2016; Uhlig et al., 2015).

2.5.3. Transferability

The in-house sensitivity assessment described in Section 2.5.2 was performed by an external laboratory, being the Laboratoire SCL de Strasbourg (Strasbourg, France) (Table S7). The real-time PCR assay was carried out using oligonucleotides (Eurogentec, Liège, Belgium), 1× SsoAdvanced universal probes supermix (Bio-Rad, Hercules, USA) and CFX Duet Real-Time PCR System (Bio-Rad, Hercules, USA).

2.5.4. Applicability

The developed real-time PCR method was tested on several commercial herbal weight-loss supplement products (samples n°1–10). For each product, 25 ng of DNA, prepared as described in Section 2.2, was tested in duplicate. All samples, collected on the European market, were either labelled or suspected to contain *Hoodia* ingredient (Table S8). These samples were also previously analyzed for the presence or absence of *Hoodia* ingredient by chemical analysis by using liquid chromatography coupled to mass spectrometry (LC-MS), allowing to evaluate the consistency and robustness of the developed real-time PCR method across the diverse matrices tested (Table S8). Moreover, these samples were tested using a plant-specific real-time PCR method as previously described (Table S8).

3. Results and discussion

3.1. In-house ITS2 sequence generation and in-silico real-time PCR design

Given the limited publicly available sequence data for *Hoodia* species, the ITS2 region was selected to develop the taxon-specific real-time PCR method for detecting *Hoodia* DNA. The ITS2 region is a molecular marker characterized by high sequence variability, offering discriminatory power for differentiating plant species and genera. This approach

has been successfully applied in previous studies to develop real-time PCR methods targeting other underrepresented or taxonomically complex plant groups, such as *Ephedra* and *Panax* (Lou et al., 2022; Zheng et al., 2021). A search of public sequences databases yielded only 16 ITS2 sequences belonging to the *Hoodia* genus and covering 5 different species including *H. officinalis*, *H. mossamedensis*, *H. gordonii*, *H. juttae* and *H. alstonii* (Fig. 1A; Table S3B). Therefore, to strengthen this dataset, additional ITS2 sequences were generated in-house from 13 different *Hoodia* plants collected from various sources. These included 2 specimens of *H. currorrii*, 4 specimens of *H. gordonii*, 3 specimens of *H. parviflora* and one specimen of *H. juttae*, *H. pedicellate*, *H. pilifera* and *H. ruschii* (Fig. 1A; Tables S1, S3A).

The resulting ITS2 *Hoodia* dataset, comprising 29 sequences, was then used for *in-silico* real-time PCR assay design. A set of two primers (*Hoodia*-F; *Hoodia*-R) and one probe (*Hoodia*-P) was initially designed to specifically amplify a 88 bp fragment of the *Hoodia* ITS2 region (Fig. 1A; Tables S3, S9). The *in-silico* method's inclusivity was confirmed using all available *Hoodia* ITS2 sequences, obtained from both public databases and in-house sequencing efforts. For all these sequences, a match of 100% in terms of identity and coverage was observed with the expected PCR amplicon sequence from the developed *Hoodia* real-time PCR method (Fig. 1A; Table S3). In addition, the *in-silico* method's exclusivity was established by blasting the expected PCR amplicon sequence against the NCBI nucleotide collection (nr/nt). No sequence outside *Hoodia* showed a 100% match in identity and coverage (Table S4A).

Although the method specificity was successfully assessed *in-silico*, the high sequence similarity observed within the targeted ITS2 region between *Hoodia* and closely related species raised concerns about potential cross-hybridizations and false-positive amplifications (Fig. 1B; Table S4B). In particular, within the *Hoodia*-P probe sequence, 10 different clusters of sequence variations were identified among closely related species when compared to the consensus *Hoodia* PCR amplicon sequence. Each variation cluster differed from *Hoodia* by one or few single-nucleotide polymorphisms (SNPs). Consequently, for effectively discriminating target and non-target sequences, the initial set of two primers and one probe was supplemented by two additional probes (*non-Hoodia*-Pa; *non-Hoodia*-Pb) designed based on single-nucleotide variations identified within the *Hoodia*-P probe sequence, for increased probe hybridization competition (Fig. 1B; Table S4C). Furthermore, LNA bases were incorporated into all probes to enhance specific hybridization efficiency. The performance of the developed *Hoodia* real-time PCR method was then experimentally assessed according an international standard guidance document, being the MPR for GMO analysis of ENGL (Marchesi et al., 2015).

3.2. Experimental performance assessment of the developed real-time PCR method targeting *Hoodia*

3.2.1. Specificity

Following *in-silico* investigations, the specificity of the developed *Hoodia* real-time PCR method was then experimentally tested using DNA extracted from a range of several target and non-target materials (Table 1; Tables S1–2). This is crucial to ensure a harmonized and reliable control by all enforcement laboratories.

As target materials, a total of 13 *Hoodia* DNA samples, covering 7 different *Hoodia* species, were tested. For all these samples, as expected, a PCR amplification signal was detected with the *Hoodia*-P probe (Table 1; Tables S1–2, S10). Moreover, the PCR amplicon sequence generated experimentally using the developed *Hoodia* real-time PCR method matched the expected reference sequence (Table S6). Notably, variation in C_q values was observed among the *Hoodia* samples, despite testing each at comparable estimated haploid genome copy numbers (Table 1). This variability is likely attributable to the use of the ITS2 region as the assay target in the proposed taxon-specific real-time method. This multi-copy nuclear marker is known to exhibit both inter-

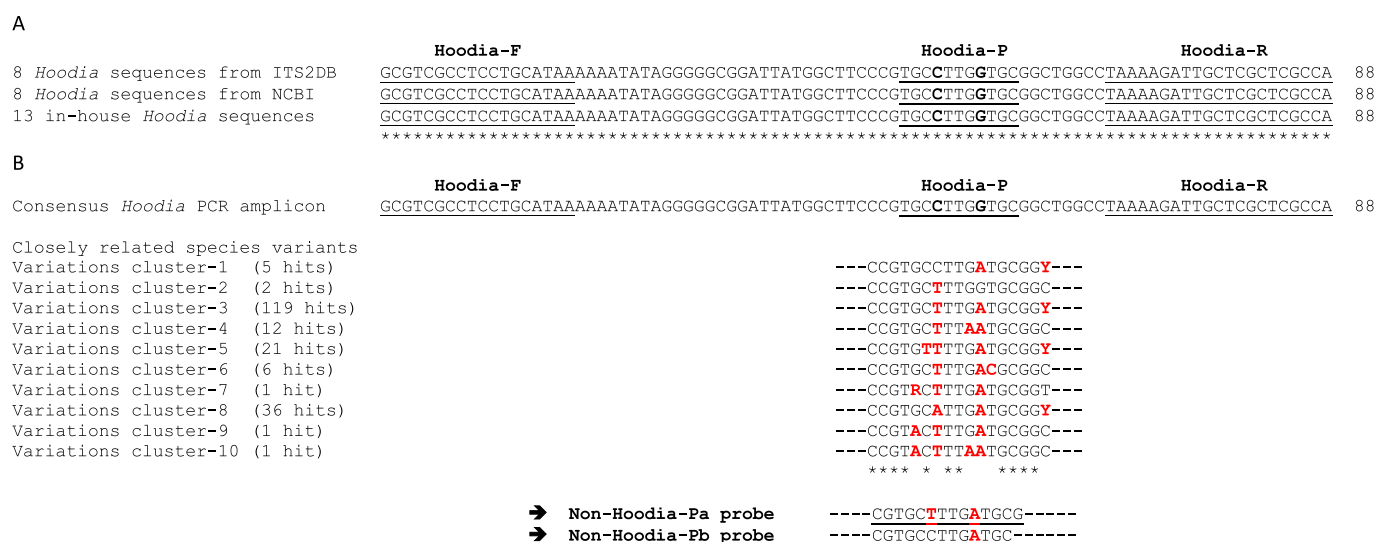


Fig. 1. Oligonucleotides design. (A) Sequence alignment of the 88 bp *Hoodia* PCR amplicon using all 29 available *Hoodia* ITS2 region sequences, including 8 from the vascular plant ITS2 database, 8 from the NCBI database and 13 generated in-house (Table S3). The positions of the designed oligonucleotide (Hoodia-F, Hoodia-R and Hoodia-P) are underlined. (B) Sequence alignment at the Hoodia-P probe binding site between the consensus *Hoodia* PCR amplicon sequence (derived from Fig. 1A) and all observed sequence variations from closely related species (Clusters 1–10) (Table S4). For each variation cluster, the number of associated hits from closely related species is indicated in parentheses (Table S4). All observed single-nucleotide variations (SNVs) are shown in red. SNVs indicated in grey were used to design the two non-Hoodia probes (Non-Hoodia-Pa and Non-Hoodia-Pb). The positions of all designed oligonucleotides are underlined.

and intra-specific variation, including sequence heterogeneity among rDNA repeats and differences in rDNA copy number (Gu et al., 2013; Liu et al., 2023; Saini et al., 2008; Song et al., 2012; Wang, Zhang, et al., 2023; Yao et al., 2010).

As non-target materials, a diverse set of DNA samples was tested in order to assess the real-time PCR method ability to discriminate *Hoodia* DNA from a wide range of non-targets, including species taxonomically related to *Hoodia*, as well as species commonly found in herbal weight-loss food products (Table 1; Tables S1–2). First, the panel comprised 17 DNA samples from plant species closely related to *Hoodia*, including *Caralluma* genus which is well-known for being used in herbal weight-loss food products. Second, 10 additional DNA samples from plant species not closely related to *Hoodia* but commonly used in herbal weight-loss food products were included, such as *C. aurantium* (bitter orange), *C. annuum* (paprika), *C. papaya* (papaya), *C. scolymus* (artichoke), *C. sinensis* (tea plant), *F. vulgare* (fennel), *G. cambogia*, *I. paraguayensis* (Yerba mate), *O. ficus indica* (Prickly pear cactus) and *P. cupana* (Guanara). Finally, this set also comprised 10 DNA samples from plant species neither closely related to *Hoodia* nor commonly observed in herbal weight-loss food products, as well as 4 non-plant DNA samples belonging to animal and microbial species. For all of these 41 negative materials, as expected, no PCR amplification signal was detected with the Hoodia-P probe (Table 1; Tables S1–2).

Based on all these results, the absence of false-positive and false-negative signals was demonstrated. No unexpected PCR amplification was observed, even with species closely related to *Hoodia* or commonly used in herbal weight-loss food products. Consequently, the developed *Hoodia* real-time PCR method was assessed as specific to target *Hoodia* DNA.

3.2.2. Sensitivity

The method's sensitivity was experimentally evaluated in accordance with international standard guidance documents (Grohmann et al., 2016; Marchesi et al., 2015; Uhlig et al., 2015). This assessment required determination of the limit of detection LOD_{95%}, defined as the lowest number of target DNA sequence copies that can be detected with a 95% probability. To this end, a range of different estimated copy numbers (25, 20, 15, 10, 5, 1, 0.1 and 0) from an artificial control carrying a single copy of the 88 bp sequence fragment of the *Hoodia* ITS2 region

targeted by the developed *Hoodia* real-time PCR method was used (Table 2; Tables S3B–C).

A PCR amplification signal was observed at levels as low as 1 estimated target copy, and at 10 estimated target copies for all 12 replicates. According to POD (probability of detection) modelling, the limit of detection LOD_{95%}, was determined at ~10 estimated target copies (LOD_{95%} = 9.166, with a 95% confidence interval of [5.908; 14.227]). Plausibility check revealed no irregularities (Table S7) (Grohmann et al., 2016; Uhlig et al., 2015). With an LOD_{95%} below 25 estimated target copies, the developed *Hoodia* real-time PCR method was evaluated as sensitive and compliant with the MPR guidance document defined by ENGL (Marchesi et al., 2015). This is important because *Hoodia* DNA may be present in small amounts in the final commercial products, especially when mixed with other ingredients. Additionally, it is noteworthy that this sensitivity assessment was conducted using an artificial control carrying a single copy of the targeted *Hoodia* ITS2 region. This approach allows precise control of the number of target copies tested, regardless of the target copy number in the genome. In plant genomes, the ITS2 region typically occurs in multiple copies, which further increases the likelihood of successful real-time PCR detection (Yao et al., 2010).

3.2.3. Transferability

To evaluate the transferability of the in-house validated *Hoodia* real-time PCR method, the same experimental setup used for the in-house sensitivity testing was applied by an external laboratory. These results closely aligned with those from the in-house validation (Table 2; Table S7). Specifically, a PCR amplification signal was detected as low as 1 estimated target copies, and at 5 estimated target copies for all 12 replicates. Additionally, the plausibility check indicated no irregularities and the LOD_{95%} was calculated at ~6 estimated target copies (LOD_{95%} = 5.222, with a 95% confidence interval of [3.044; 8.943]). The consistency between in-house and external laboratory results confirms the transferability of the *Hoodia* real-time PCR method.

3.2.4. Applicability

The method's applicability was experimentally evaluated to determine whether it can be reliably applied to commercial herbal dietary supplements, considering the complexity of food matrices and

Table 1
Experimental specificity assessment of the *Hoodia* real-time PCR method using different target and non-target DNA materials. If applicable, the common name for the species is provided while “n.a.” is indicated if not applicable. The presence and absence of amplification are respectively denoted by “+” and “-”. Each result was obtained from duplicate PCR reactions conducted on about 3000 estimated haploid genome copies. The means of the measured C_q values and associated standard deviations are provided in brackets. When multiple plant specimens belonging to the same species are tested, each is differentiated by a number within parentheses next to the plant's name. Plant species commonly found in herbal weight-loss dietary products are marked with an asterisk, and those reported to be combined with *Hoodia* ingredients in such products are marked with “Ψ” (Cohen et al., 2023; Corley & Miller, 2002; de Freitas Junior & de Almeida, 2017; Dutt et al., 2012; Elvir-Lazo et al., 2024; Gamboa-Gómez et al., 2015; König, 2024; Pittler et al., 2005; Sirotkin, 2022; Sun et al., 2021).

Kingdom	Family	Species	Common name	<i>Hoodia</i> real-time PCR assay
Animalia	Hominidae	<i>Homo sapiens</i>	Human	–
Bacteria	Bacillaceae	<i>Bacillus licheniformis</i>	n.a.	–
Fungi	Trichocomaceae	<i>Aspergillus niger</i>	n.a.	–
	Pichiaceae	<i>Pichia pastori</i>	n.a.	–
Plantae	Amaranthaceae	<i>Beta vulgaris</i>	Sugar beet	–
	Apiaceae	<i>Foeniculum vulgare</i> * Ψ	Fennel	–
	Apocynaceae	<i>Caralluma europea</i> * Ψ	n.a.	–
		<i>Caralluma europaea</i> var. <i>Europea</i> * Ψ	n.a.	–
		<i>Caralluma joannis</i> * Ψ	n.a.	–
		<i>Caralluma hesperidum</i> * Ψ	n.a.	–
		<i>Ceropegia woodii</i>	String of hearts	–
		<i>Dischidia nummularia</i>	Button orchid	–
		<i>Dischidia ruscifolia</i>	Million hearts plant	–
		<i>Dischidia</i> sp.	n.a.	–
		<i>Hoodia currorrii</i> * (specimen n° 1)	n.a.	+ (C _q : 22.3 ± 0.1)
		<i>Hoodia currorrii</i> * (specimen n° 2)	n.a.	+ (C _q : 20.3 ± 0.0)
		<i>Hoodia gordonii</i> * (specimen n° 1)	n.a.	+ (C _q : 19.1 ± 0.0)
		<i>Hoodia gordonii</i> * (specimen n° 2)	n.a.	+ (C _q : 21.8 ± 0.0)
		<i>Hoodia gordonii</i> * (specimen n° 3)	n.a.	+ (C _q : 19.0 ± 0.0)
		<i>Hoodia gordonii</i> * (specimen n° 4)	n.a.	+ (C _q : 19.2 ± 0.1)
		<i>Hoodia juttae</i> *	n.a.	+ (C _q : 16.6 ± 0.0)
		<i>Hoodia parviflora</i> * (specimen n° 1)	n.a.	+ (C _q : 19.3 ± 0.0)
		<i>Hoodia parviflora</i> * (specimen n° 2)	n.a.	+ (C _q : 21.3 ± 0.0)
		<i>Hoodia parviflora</i> * (specimen n° 3)	n.a.	+ (C _q : 25.9 ± 0.0)
		<i>Hoodia pedicellate</i> *	n.a.	+ (C _q : 20.0 ± 0.1)

Table 1 (continued)

Kingdom	Family	Species	Common name	<i>Hoodia</i> real-time PCR assay
		<i>Hoodia pilifera</i> *	n.a.	+ (C _q : 19.8 ± 0.1)
		<i>Hoodia ruschii</i> *	n.a.	+ (C _q : 15.8 ± 0.1)
		<i>Hoya multiflora</i>	Shooting star hoyo	–
		<i>Huernia keniensis</i>	n.a.	–
		<i>Huernia rubra</i>	n.a.	–
		<i>Orbea variegata</i>	Starfish plant	–
		<i>Rauvolfia vomitoria</i>	Poison devil's-pepper	–
		<i>Senna alexandrina</i>	n.a.	–
		<i>Stapelia flaviviridis</i> / <i>grandiflora</i>	Carrión flower	–
		<i>Stapelia tromotriche</i> <i>revoluta</i>	Carrión flower	–
		<i>Strophanthus amboensis</i> / <i>petersianus</i>	n.a.	–
	AQUIFOLIACEAE	<i>Ilex paraguariensis</i> * Ψ	Yerba mate	–
	ASTERACEAE	<i>Cynara scolymus</i> *	Artichoke	–
	BRASSICACEAE	<i>Brassica napus</i>	Rapeseed	–
	CACTACEAE	<i>Opuntia ficus indica</i> * Ψ	Prickly pear cactus	–
	CARICACEAE	<i>Carica papaya</i> *	Papaya	–
	CLUSIACEAE	<i>Garcinia cambogia</i> / <i>gummi-gutta</i> * Ψ	n.a.	–
	FABACEAE	<i>Glycine max</i>	Soybean	–
	MALVACEAE	<i>Gossypium hirsutum</i>	Cotton	–
	POACEAE	<i>Oryza sativa</i>	Rice	–
		<i>Triticum aestivum</i>	Wheat	–
		<i>Zea mays</i>	Maize	–
	RUTACEAE	<i>Citrus aurantium</i> * Ψ	Bitter orange	–
	SAPINDACEAE	<i>Paullinia cupana</i> * Ψ	Guarana	–
	SOLANACEAE	<i>Capsicum annuum</i> * Ψ	Paprika	–
		<i>Nicotiana tabacum</i>	Tobacco	–
		<i>Solanum lycopersicum</i>	Tomato	–
		<i>Solanum tuberosum</i>	Potato	–
	THEACEAE	<i>Camellia sinensis</i> * Ψ	Tea plant	–

associated challenges (i.e., DNA degradation), which is crucial for enforcement purposes. The developed *Hoodia* real-time PCR method was applied to 10 different commercial herbal weight-loss food supplement products collected from the European market. These products were either purchased from e-commerce platforms and regular shops or sampled by competent authorities as part of their national control plans (Table 3; Table S8). All these samples were labelled as containing *Hoodia* ingredient, except for samples n°9–10. These two samples were suspected to contain *Hoodia* based on recent findings using multidimensional chromatographic fingerprinting combined with chemometrics (Ranjan et al., 2023). Furthermore, previous chemical-based analysis of all samples using LC-MS detected the presence of *Hoodia* ingredients in samples n°1–4, providing a reference for evaluating the consistency and robustness of the developed real-time PCR method across the diverse matrices tested (Table 3; Table S8).

First, using the developed real-time PCR method, the presence of *Hoodia* ingredients was detected in samples n°1–5, which is consistent with the labelling of these products (Table 3). These real-time PCR results were also largely in agreement with the LC-MS data, except for sample n°5 (Table 3; Table S8). This discrepancy observed for sample n°5 could however be explained by the low amount of *Hoodia* present, as

Table 2

Experimental sensitivity assessment of the *Hoodia* real-time PCR method for the in house validation and transferability assays. The presence and absence of amplification are respectively denoted by “+” and “-”. Each result was obtained from 12 replicate PCR reactions. The number of positive replicate(s) out of the 12 replicates tested is indicated in brackets. The means of the measured C_q values and associated standard deviations are provided in brackets.

	Estimated target copy number							
	25	20	15	10	5	1	0.1	0
In house real-time PCR validation assays	+	+	+	+	+	+	–	–
	(12/12)	(12/12)	(12/12)	(12/12)	(9/12)	(3/12)	(0/12)	(0/12)
Transferability real-time PCR assays	(C_q : 34.2 ± 0.4)	(C_q : 34.4 ± 0.2)	(C_q : 35.0 ± 0.5)	(C_q : 35.4 ± 0.5)	(C_q : 36.4 ± 0.5)	(C_q : 37.6 ± 0.1)	–	–
	+	+	+	+	+	+	–	–
	(12/12)	(12/12)	(12/12)	(12/12)	(12/12)	(4/12)	(0/12)	(0/12)
	(C_q : 33.6 ± 0.3)	(C_q : 34.0 ± 0.6)	(C_q : 34.2 ± 0.4)	(C_q : 35.2 ± 0.6)	(C_q : 36.3 ± 0.8)	(C_q : 37.7 ± 0.7)		

Table 3

Experimental applicability assessment of the *Hoodia* real-time PCR method using different commercial herbal weight-loss supplement products (samples n°1–10). Available labelling information is provided (Table S8). All samples, except n°9–10, listed *Hoodia* as an ingredient on the product label. For each sample, 25 ng of extracted DNA was tested in duplicate. Positive and negative PCR signals are respectively symbolized by “+” or “-”. The means of measured C_q values and associated standard deviations are indicated in brackets. The presence of previously detected plant DNA by real-time PCR in these samples is indicated by an asterisk. In addition, the presence of *Hoodia* ingredients previously detected in these samples by chemical-based analysis (LC-MS) is indicated by “Ψ” (Table S8).

N°	Sample description	Product formulation	<i>Hoodia</i> real-time PCR assay
1*Ψ	<i>Hoodia</i> nutritional supplement – <i>Hoodia gordonii</i> powder	Solid - Capsule	+
			(C_q : 17.0 ± 0.1)
2*Ψ	<i>Hoodia</i> dietary supplement – <i>Hoodia gordonii</i> powder	Solid - Capsule	+
			(C_q : 25.8 ± 0.1)
3*Ψ	<i>Hoodia</i> dietary supplement – <i>Hoodia gordonii</i> powder	Solid - Capsule	+
			(C_q : 29.0 ± 0.0)
4*Ψ	Metabolism support - <i>Hoodia gordonii</i> powder	Solid - Capsule	+
			(C_q : 32.9 ± 0.0)
5*	<i>Hoodia</i> dietary supplement – <i>Hoodia gordonii</i> powder	Solid - Tablet	+
			(C_q : 33.1 ± 0.1)
6	Hunger control-herbal supplement – <i>Hoodia</i> root	Solid - Capsule	–
7	Metabolism booster-herbal supplement - <i>Hoodia</i>	Solid - Capsule	–
8*	Weight loss supplement appetite suppressant - <i>Hoodia</i>	Solid - Capsule	–
9	Fat burning – controls appetite - promotes weight loss	Solid - Capsule	–
10	Weight loss formula	Solid - Capsule	–

suggested by the high C_q value obtained by real-time PCR, indicating levels probably near or below of the detection limit of the chemical-based method (Table 3; Table S7). This finding highlighted the added value of using the real-time PCR method for detecting trace amounts of *Hoodia* ingredients. Second, although the product labelling indicated the presence of *Hoodia*, neither the real-time PCR nor LC-MS methods detected any *Hoodia* ingredient in samples n°6–8. This suggests that the *Hoodia* ingredient is either present at trace level below the limit of detection of these methods or that mislabelling occurred (Table 3; Table S8). Finally, as expected, the presence of *Hoodia* was not observed by either the real-time PCR or LC-MS methods in sample n°9–10, which were not labelled as containing *Hoodia* (Table 3; Table S8). Taken together, these results demonstrate the applicability of the developed *Hoodia* real-time PCR method for reliable detection of *Hoodia* DNA across a range of commercial herbal weight-loss supplement products.

As a complementary investigation, all samples were also tested in parallel using a universal plant real-time PCR assay (Tables 3; S8). Amplifiable plant DNA was observed in all samples where *Hoodia* DNA was detected (n°1–5) as well as in sample n°8 where not *Hoodia* DNA was detected. No amplifiable plant DNA was detected in samples n°6–7

and 9–10 where consistently no *Hoodia* DNA was detected. These results transparently illustrate the variability encountered when monitoring processed herbal products and emphasize the importance of prior DNA quality assessment to ensure reliable data interpretation.

4. Conclusion

In this study, a *Hoodia* real-time PCR method was developed for the first time and validated according a widely recognized international standard for real-time PCR method performance (Marchesi et al., 2015). The assay targets ITS2, a well-established molecular marker for plant authentication (Banchi et al., 2020; Frigerio et al., 2021; Gu et al., 2013; Yao et al., 2010). This real-time PCR methods was specifically designed for genus-level detection of *Hoodia*, in line with CITES regulatory requirements and the practical needs of enforcement laboratories. The method's development was strongly supported by prior in-house sequencing efforts of the *Hoodia* ITS2 region, providing essential reference data for designing specific primers and probes capable of distinguishing the *Hoodia* genus from closely related taxa.

This novel real-time PCR method demonstrated high specificity for *Hoodia* DNA, with no PCR amplification signal observed for non-*Hoodia* species, including closely related ones and other species commonly found in commercial herbal weight loss products. This method also exhibited high sensitivity, with a limit of detection below 25 estimated target copies ($LOD_{95\%} \sim 10$). Furthermore, the transferability of this in-house validated method to an external laboratory was confirmed, with consistent results between in-house and external assays. Finally, using 10 different commercial herbal weight-loss supplements, the applicability of this method was also successfully tested, where it reliably detected *Hoodia* DNA, even at trace levels. For enforcement laboratories, this real-time PCR method directly addresses the analytical challenges posed by marketed herbal weight-loss products, including complex processed mixtures of different plant species. Collectively, these results demonstrate the method's robustness and practical applicability, laying the groundwork for future efforts such as an international collaborative ring trial involving multiple enforcement laboratories to further strengthen confidence in method performance and promote harmonization of *Hoodia* testing across laboratories (Broeders et al., 2014; Marchesi et al., 2015).

This study offers enforcement laboratories a successfully validated real-time PCR method, which remains the gold standard for DNA-based taxon detection. This user-friendly key molecular tool is a valuable resource for official controls, helping to ensure sustainability, authenticity, traceability, quality and safety of commercial *Hoodia*-based products.

CCRediT authorship contribution statement

Marie-Alice Fraiture: Writing – review & editing, Writing – original draft, Validation, Supervision, Methodology, Investigation, Formal analysis, Conceptualization. **Andrea Gobbo:** Writing – review & editing,

Validation, Formal analysis. **Patrick Philipp**: Writing – review & editing, Formal analysis. **Nina Papazova**: Writing – review & editing, Validation, Formal analysis. **Céline Vanhee**: Writing – review & editing, Supervision, Funding acquisition, Formal analysis, Conceptualization. **Nancy H.C. Roosens**: Writing – review & editing, Supervision, Funding acquisition, Conceptualization.

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Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

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Appendix A. Supplementary data

Supplementary data to this article can be found online at <https://doi.org/10.1016/j.fochms.2026.100367>.

Data availability

Data will be made available on request.

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